

Paulina Matkowska

Data Science & Bioprocess Scientist

About Me

Bioprocess and Data Scientist with 5+ years of experience across GMP- and GxP-regulated biopharma, biotech manufacturing, and quality control, holding an MSc in Data Science. I transform complex process data into contextualized, analysis-ready data through robust metadata management, taxonomies, ontologies, and standardized multi-vendor pipelines across upstream operations.

A passionate and agile scientist driven by innovation, digital transformation, and data governance best practices. Specializing in data stewardship and bridging modern bioprocessing with data automation to drive process quality and development. Eager to leverage strong cross-functional collaboration and inspection readiness standards within a multidisciplinary team.

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Education

B. Sc. in Biotechnology

MANNHEIM UNIVERSITY OF
APPLIED SCIENCES, DE
2018 – 2022

MSC DATA SCIENCE

UNIVERSITY OF ESSEX, UK
2023 – 2026

Experience

BIOPROCESS SCIENTIST & DATA LEAD

08.2024–

The Cultured Hub | Startup Environment

Present

- Designed and implemented unified metadata models and data taxonomies for process data across multiple bioreactor systems and instrument types from diverse vendors.
- Built and deployed interactive Power BI dashboards integrated with Python scripts, transforming disparate sensor inputs into contextualized, analysis-ready data and automating cross-system data pipelines.
- Standardized and synchronized disparate data streams coming from diverse stainless steel/glass bioreactors, sensors, and lab equipment, handling complex multi-vendor data formats.
- Executed hands-on bioprocess development and daily operations using diverse stainless steel and glass bioreactors, sensors, and lab equipment, bridging pharma and food industry standards.
- Authored SOPs and critical documentation under strict GDP and GLP standards, driving digital manufacturing workflows and maintaining full audit and inspection readiness.

SENIOR TECHNICIAN GMP OPERATIONS

10.2022–

Janssen Vaccines by J&J

07.2024

- Executed upstream and downstream project work in a regulated cGMP environment.
- Organized and optimized pilot plant facilities according to safety guidelines and cGMP standards, driving process data accuracy and operational reliability.
- Investigated non-conformances, identified root causes, and implemented corrective actions (CAPAs) in strict adherence to cGMP quality frameworks.
- Provided theoretical and practical training (instructor-led courses) on critical documentation, equipment operation, and manufacturing processes.
- Managed downstream (DSP) process planning and team qualification planning.

MSc THESIS PROJECT: MACHINE LEARNING IN BIOPROCESS MODELING

- Developed artificial neural networks (ANNs) and transfer learning frameworks to optimize bioreactor scale-up strategies and predict cell culture process parameters.
- Processed multi-dimensional biological datasets in Python to build predictive models, bridging advanced data science with scalable bioprocess engineering.

CORE SKILLS & EXPERTISE

- Data Architecture & Governance: Metadata management, taxonomies, ontologies, data stewardship, and data governance frameworks.
- Data Engineering & Analytics: Automated multi-vendor data pipelines, Python, Power BI, and transformation into contextualized, analysis-ready data.
- Bioprocess Development: Upstream and downstream processing, bioreactor scale-up, sensor integration, and digital manufacturing platforms.
- Regulatory & Quality Compliance: GMP, GxP, GDP, GLP, strict adherence to SOPs, and full audit / inspection readiness.

SOFT SKILLS

- Cross-Functional Communication: Proven ability to bridge the gap between technical, scientific, and business teams, driving effective collaboration across multidisciplinary environments.
- Analytical Problem-Solving: Strong critical thinking focused on translating complex manufacturing and data challenges into structured, scalable operational solutions.
- Adaptability & Continuous Learning: Highly adaptable to fast-paced startup and enterprise environments, demonstrating a growth mindset and a passion for digital transformation.
- Autonomy & Project Ownership: Self-disciplined and capable of managing complex, research-oriented tasks from conception to execution with minimal oversight.
- Knowledge Sharing & Leadership: Experienced in delivering technical training, authoring clear documentation, and promoting data-driven ways of working across teams.

LANGUAGES

- English
- German
- Polish

BACHELOR THESIS PROJECT (DIVISION OF FUNCTIONAL GENOME ANALYSIS)

02.2022 –
05.2022

DKFZ Deutsches Krebsforschungszentrum

- Designed, conducted, and interpreted molecular experiments, investigating cell function, gene expression, and signaling mechanisms.
- Analyzed data using Excel, R, FlowJo, and GraphPad Prism, collaborating with interdisciplinary scientists.
- Thesis: "Role of ATG7 and UBE2T in the Progression of Pancreatic Cancer" (Grade 1.0)
- GPA 1.9

INTERNSHIP MICROBIOLOGY QUALITY CONTROL

05.2021 –
11.2021

WESSLING GmbH

- Conducted comprehensive microbiological analyses on food and water samples to ensure safety, quality compliance, and process hygiene.
- Identified sources of microbial contamination in close collaboration with food manufacturers, utilizing data-driven tracking to improve facility sanitation protocols.
- Researched food-borne pathogens (including Salmonella, E. coli, and Staph infections) to support risk assessments and quality assurance measures.

QUALITY ASSURANCE

10.2017 –
08.2018

Star Produktions GmbH

- Performed precise physicochemical analyses of raw materials and finished products to enforce strict quality compliance standards.
- Developed and validated innovative test methods to enhance product testing accuracy and laboratory efficiency.
- Managed laboratory operations and process control activities for cosmetics and cleaning product manufacturing.
- Maintained meticulous GMP and quality documentation to meet rigorous industry standards and ensure audit-readiness.